Proofs with Sets

$A = \{0, 1, 2, 3, 4, 5\}$ and $R = \{(1, 2), (3, 3), (2, 5), (4, 2), (4, 3), (5, 0)\}$

Definition 1. *A relation R on $\mathbf{X}$ is **reflexive** if for all $a \in \mathbf{X}, aRa$*

Definition 2. *A relation R on $\mathbf{X}$ is **symmetric** if for all $a, b \in \mathbf{X}$, if aRb then bRa.*

Definition 3. *A relation R on $\mathbf{X}$ is **transitive** if for all $a, b, c \in \mathbf{X}$, if aRb and bRc, then aRc.*

Is R reflexive? Explain.

Proof. Say we choose $m = 2$.

Since $gcd(2, 2) = 2 \neq 1$, it follows that mRm is false.

Thus R is not reflexive □

Is R symmetric? Explain.

Proof. Choose arbitrary natural numbers a and b.

Suppose aRb.

Then, we have that $gcd(a, b) = 1$.

Thus, $gcd(b, a) = 1$.

So bRa.

Therefore, we conclude that R is symmetric. □

Is R transitive? Explain.

Proof. Consider $2, 3, 4 \in \mathbb{N}$.

Since $gcd(2, 3) = 1$, it follows that $2R3$. Since $gcd(3, 4) = 1$, it follows that $3R$.

But, $gcd(2, 4) = 2 \neq 1$. So $2 \not R$. □